Student Requirements Specification

for

Student Management System, Release 1.0

Version 1.0 approved

Prepared by Team 1

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Revision History

Name	Date	Reason For Changes	Version
Đỗ Đăng Khoa	10/03/2026	Created initial SRS outline; drafted Introduction (Purpose, Scope, References); set up document structure and formatting.	1.0 draft 1
Cao Tiến Phát	13/03/2026	Authored Overall Description: Product Perspective, User Classes, Operating Environment, Constraints, Assumptions/Dependencies.	1.0 draft 1
Trần Việt Long	16/03/2026	Drafted System Features section: Course Registration and Input Grades (description + functional requirements tables).	1.0 draft 2
Đỗ Đăng Khoa	20/03/2026	Added Data Requirements: logical data model overview and initial data dictionary entries; aligned terminology and naming conventions.	1.0 draft 3
Cao Tiến Phát	24/03/2026	Completed Reports section (SMS-RPT-1 to SMS-RPT-6) and updated related data integrity notes.	1.0 draft 4
Trần Việt Long Cao Tiến Phát Đỗ Đăng Khoa	29/03/2026	Final review and consolidation: polished wording, ensured consistency across sections, verified TOC entries, and prepared Version 1.0 for approval.	1.0 approved

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1. Introduction

1.1 Purpose

This SRS document describes the functional and non-functional requirements for version 1.0 of the Student Management System (SMS). This document is intended for the development and testing teams to ensure the system operates as designed.

1.2 Document Conventions

No special typographical conventions are used in this SRS.

1.3 Project Scope and Product Features

The SMS system will allow university students and faculty to perform online academic management tasks, including course registration, grade entry, and progress tracking. Detailed features planned for this version include profile management, course registration, and academic performance reporting.

1.4 References

1. *Student Management System – Vision and Scope Document, Internal Project Document, Team 1, 2026.*

2. *Student Management System – Use Cases Document, Internal Project Document, Team 1, 2026.*

3. *Student Management System – Business Rules Document, Internal Project Document, Team 1, 2026.*

4. *IEEE Std 29148-2018, Systems and Software Engineering — Life Cycle Processes — Requirements Engineering, IEEE.*

2. Overall Discription

2.1 Product Perspective

The Student Management System is a new software system that replaces the current paper-based student management process and manual spreadsheets. The system is expected to evolve through several versions, eventually connected with an online tuition payment system and a mobile application for parents.

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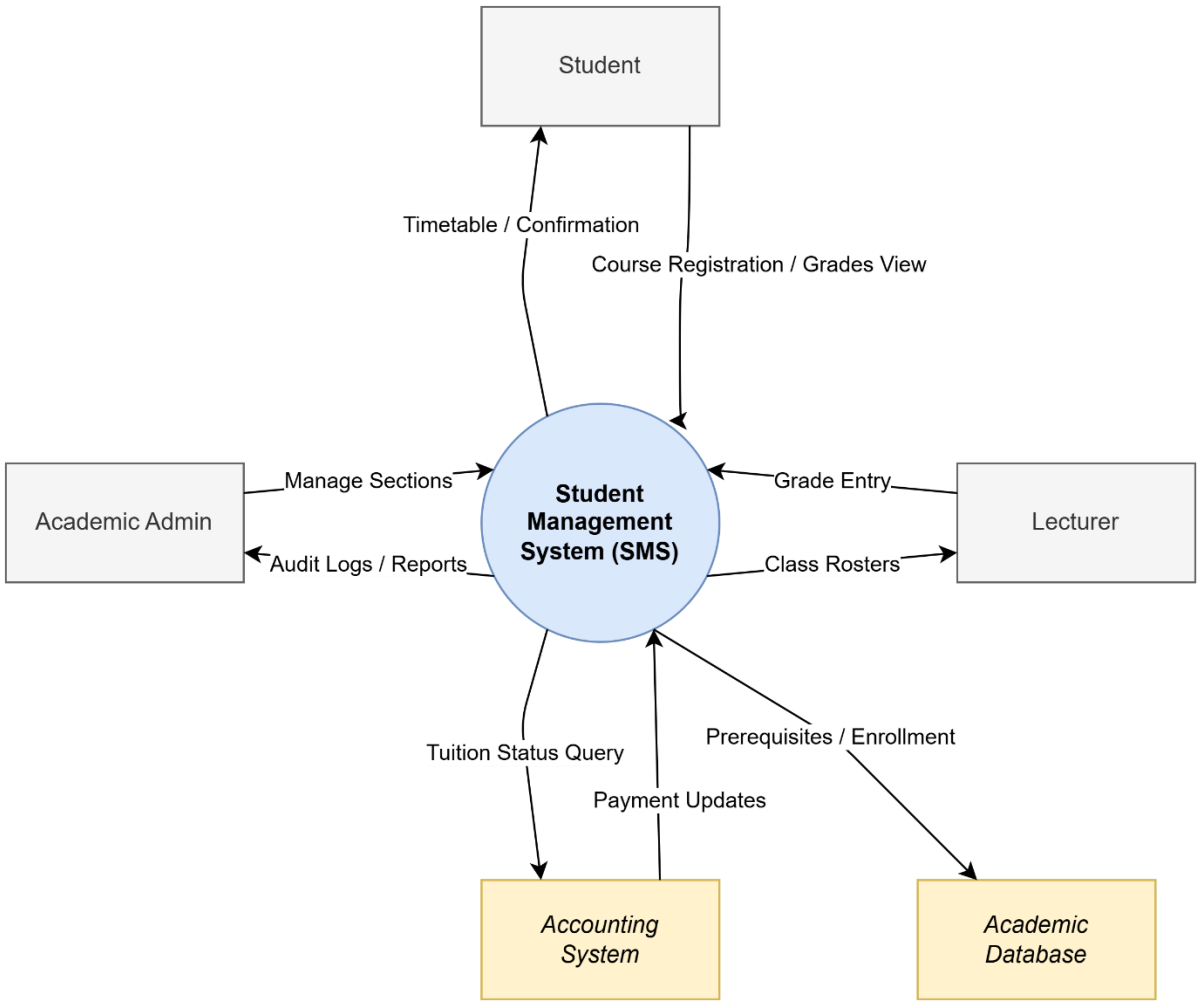

Figure 1. Context diagram for release 1.0 of the Student Management System.

2.2 User Classes and Characteristics

Student (Primary User Group)	Students use the SMS to perform personal academic tasks such as: registering for courses, viewing timetables, checking grades, and paying tuition fees. There are approximately 10,000 active students. On average, each student will access the system 3 to 5 times per week. However, the system must specifically handle extreme **peak loads** during the initial course registration periods of each semester. An estimated 70% of students will interact with the system via the Mobile App, while the remaining 30% will use the Web Portal.
Lecturer	The university employs about 500 Lecturers. They will use the SMS to receive class rosters, input midterm/final grades, take attendance, and communicate academic information. System usage frequency will spike dramatically during the final grading period. Due to the critical nature of grading data, the interface designed for Lecturers requires high **Data Integrity** mechanisms and comprehensive User Manuals to prevent data entry errors.

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Academic Admin	This group (approximately 30 staff members) is responsible for establishing and maintaining the core "rules" of the system. They have the authority to open course sections, schedule timetables, manage the core curriculum, and update students' academic statuses. Because this group possesses high-level privileges, every Create/Read/Update/Delete (CRUD) operation they perform on the SMS must be strictly logged via an **Audit Trail** to ensure transparency and security.
Academic Advisor	When students submit academic requests (e.g., temporary leave of absence, credit overload, grade appeals), these requests are routed to the respective Academic Advisor's dashboard. Advisors will review the profiles on the SMS, approve or reject the requests, and update the processing status. They will also receive automated alerts from the system if a student under their supervision shows declining academic performance.

2.3 Operating Environment

OE-1: The SMS shall operate correctly on modern web browsers commonly used by Students and staff, including Google Chrome, Microsoft Edge, Mozilla Firefox, and Apple Safari.

OE-2: The SMS shall be deployed as a web application with a React.js front-end and a Java Spring Boot back-end.

OE-3: The SMS shall use PostgreSQL as the primary database and Redis for distributed locking during peak course registration operations.

OE-4: The SMS shall permit user access using desktop and mobile devices through the web interface.

2.4 Design and Implementation Constraints

CO-1: The system's design, code, and maintenance documentation shall conform to the *Process Impact Intranet Development Standard, Version 1.3* [2].

CO-2: The system shall use PostgreSQL as the database engine.

CO-3: The user interface shall follow responsive web design principles to support desktop and mobile browsers.

2.5 Assumptions and Dependencies

AS-1: It is assumed that all students, lecturers, academic advisors, and academic administrators have access to a stable internet connection and compatible devices (PC, laptop, or mobile) to use the Student Management System.

AS-1: It is assumed that the University's Academic Database contains accurate and up-to-date student records, course information, prerequisite mappings, and scheduling data.

DE-1: The operation of the SMS depends on real-time integration with the Accounting System to determine tuition payment status before allowing course registration.

DE-2: The operation of the SMS depends on the availability and accuracy of the Academic Database for retrieving prerequisite completion data, enrollment history, timetable information, and grade records.

DE-3: The SMS depends on the University Authentication Service for secure login, identity verification, and role-based access control. Sequence Diagrams

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3. System Features

3.1 Course Registration

3.1.1 Description

The Course Registration feature allows eligible Students to register for, drop, or change course sections for an academic semester through the Student Management System (SMS). The system validates academic eligibility, financial clearance, prerequisite satisfaction, schedule conflicts, and seat availability before confirming a registration. This feature supports peak usage during official registration and add/drop periods.
Priority: High.

3.1.2 Functional Requirements

Registration.Access: Accessing the Course Registration Module	
.Access:	The SMS shall allow only authenticated Students to access the course registration module.
.Eligibility:	The SMS shall verify that the Student's academic status and financial status (e.g., no outstanding tuition debt) are cleared before allowing registration.
.Period:	The SMS shall verify that the current date falls within the officially announced course registration or add/drop period.
.Cutoff:	If the current date is outside the officially announced registration or add/drop period, the SMS shall notify the Student and prevent registration actions.
Registration.View: Viewing Available Courses	
.Semester:	The SMS shall allow the Student to select an academic semester for registration.
.List:	The SMS shall display a list of available courses and sections for the selected semester.
.Details:	Each course section shall display the course code, course name, credits, lecturer, schedule, location, and remaining seat capacity.
Registration.Validate: Validation Rules	
.Prerequiste:	The SMS shall verify that the Student has successfully completed all prerequisite courses required for the selected course section.
.Conflict:	The SMS shall detect schedule conflicts and prevent registration if the selected section overlaps with any existing registered courses.
.CreditLimit:	The SMS shall prevent registration if adding the course causes the total registered credits to exceed the maximum limit of 24 credits for a regular semester.
Registration.Select: Selecting Course Sections	
.Add:	The SMS shall allow the Student to select one or more course sections to add to a pending registration list.
. Remove:	The SMS shall allow the Student to remove a selected course section before confirmation.
.Waitlist:	If a course section has no remaining seats but allows a waitlist, the SMS shall notify the Student and offer the option to be placed on the waitlist queue.
Registration.Confirm: Confirming Registration	

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.Summary:	The SMS shall display a summary of selected courses, total credits, and a proposed timetable before confirmation.
.Confirm:	The Student shall be able to confirm, modify, or cancel the registration request.
Registration.Commit: Completing Registration	
.Transaction:	When the Student confirms registration, the SMS shall perform the following actions as a single transaction:
.Lock:	Acquire a Redis distributed lock for the selected section to ensure seat capacity is not exceeded during high-concurrency.
.Update:	Decrease the available seat counts for each registered course section.
.Timetable:	Update the Student's personal timetable.
.Notify:	Send a confirmation message to the Student via email or system notification.
.Rollback:	If any step fails, the SMS shall roll back all changes and notify the Student of the failure reason.
Registration.Commit: Completing Registration	
.Drop:	During the officially announced add/drop period, the SMS shall allow the Student to drop a registered course.
.Change:	The SMS shall allow the Student to replace a registered course with another eligible section during the add/drop period.
.Restriction:	After the add/drop period ends, the SMS shall prevent changes and display an appropriate system message.

3.2 Input Midterm/Final Grades

3.2.1 Description

This feature enables Lecturers to enter, review, and officially submit Midterm, Process, and Final Exam grades for course sections they are assigned to teach. The system validates grade formats, computes the Final Course Grade using the official weighting formula, and locks grades after official submission. It also supports draft saving and importing grades via a pre-formatted Excel template.

3.2.2 Functional Requirements

Grades.Access: Accessing the Grade Management Module	
.Access:	The SMS shall allow only authenticated Lecturers who are officially assigned to a course section to access the grade-entry module.
.Window:	The SMS shall allow grade entry only during the official grading window (7 working days after the final exam date).
.Readonly:	If the grading window has closed, the SMS shall display the grading sheet in read-only mode and disallow grade modification.
Grades.Input: Entering Midterm, Process, and Final Exam Scores	
.Display:	The SMS shall display a structured grading sheet containing the list of enrolled students and columns for Process Grade, Midterm Grade, and Final Exam Grade.
.Format:	The SMS shall validate each score entered to ensure it is a numeric value between 0.0 and 10.0 with at most one decimal place.
.Error:	If an invalid grade is entered, the SMS shall highlight the affected cell, display

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	an error tooltip, and disable "Calculate" and "Submit" actions until corrected.
.Draft:	The Lecturer shall be able to save the current grading sheet as a draft without finalizing or calculating grades.

Grades.Import: Uploading Grade Data from Excel

.Import:	The Lecturer shall be able to upload a pre-formatted Excel file containing student grade data.
.Validate:	The SMS shall parse the Excel file and verify column headers, student IDs, and row formatting.
.Reject:	If the file contains mismatched IDs, missing fields, or formatting errors, the SMS shall reject the file and display a message specifying the problematic rows.

Grades.Calculate: Computing Final Grades

.Compute:	The SMS shall compute each Student's Final Course Grade using the formula: Final Grade = (Process Score × 10%) + (Midterm Score × 30%) + (Final Exam × 60%)
.Letter:	The SMS shall assign a letter grade according to the university's grading scale after the numeric calculation.

Grades.Submit: Submitting Official Grades

.Prompt:	When the Lecturer clicks "Submit Official Grades", the SMS shall display a confirmation prompt stating that further modifications require Academic Admin approval.
.Lock:	Once confirmed, the SMS shall store all grades, mark the section grading status as "Submitted," and lock further modifications.
.Notify:	After successful submission, the SMS shall notify Students that their grades have been published.

Grades.Audit: Ensuring Integrity and Administrative Control

.Log:	Any post-submission grade change shall require Academic Admin approval and be recorded in an audit trail.
.Readonly:	When a Lecturer views officially submitted grades, the SMS shall display them in read-only mode.

Grades.Failure: Error Handling

.Failure: If any step of grade input, calculation, or submission fails, the SMS shall roll back the grading operation and inform the Lecturer of the failure reason.

3.3 View Timetable

3.3.1 Description

This feature allows Students and Lecturers to view their personal weekly or semesterly timetable. The system ensures users can only access their own schedules and presents classes in a responsive calendar layout, with options to filter by timeframe and export.

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3.3.2 Functional Requirements

Timetable.Access: Accessing the Timetable Module	
.Access:	The SMS shall allow only authenticated users with a unique ID to access the timetable module.
.Privacy:	The SMS shall restrict timetable visibility to the requesting user's own schedule.
Timetable.View: Viewing the Personal Timetable	
.RoleAware:	The SMS shall identify whether the user is a Student or Lecturer and fetch classes accordingly (enrolled classes for Students; assigned classes for Lecturers).
.Render:	The SMS shall render the timetable in a weekly and semester view, including class code, name, time, room, and instructor.
.Empty:	If no classes are scheduled for the selected period, the SMS shall display a message indicating that no classes are scheduled.
Timetable.FilterExport: Filtering and Exporting	
.Filter:	The user shall be able to switch the displayed timeframe (e.g., another semester or a specific week).
.ExportPDF:	The SMS shall allow exporting the timetable to a PDF file.
.ExportICS:	The SMS shall allow exporting the timetable to a standard ICS calendar file for integration with calendar applications.
Timetable.Security: Security and Unauthorized Access Handling	
.Unauthorized: If a user attempts to access another person's timetable, the SMS shall block the request, log the attempt, and display an authorization error message.	

3.4 Open Course Sections

3.4.1 Description

The Open Course Sections feature enables Academic Admin users to create, publish, and manage course sections for an academic semester. This includes setting section capacity, assigning a lecturer, defining meeting times and rooms, and opening or closing sections for student registration. The system validates resource constraints (room availability, lecturer schedule conflicts) and maintains an audit trail for all section changes.
Priority: High.

3.4.2 Functional Requirements

Sections.Access: Accessing the Section Management Module	
.Access:	The SMS shall allow only authenticated Academic Admin users to access the course section management module.
.Role:	The SMS shall prevent Students and Lecturers from creating, editing, opening, or closing course sections.
.Audit:	The SMS shall record an audit log entry for each create/update/open/close action on a course section, including user ID, timestamp, and changed fields.
Sections.Create: Creating a Course Section	

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.Fields: The SMS shall require the Academic Admin to specify at minimum: Course Code, Semester, Section Number, Capacity, Lecturer, Schedule (day/time), and Room.

.Validate: The SMS shall validate that Capacity is a positive integer and that Course Code exists in the curriculum catalog.

.DefaultState: The SMS shall create new sections in a "Draft/Closed" state by default until explicitly opened for registration.

Sections.Validate: Conflict and Resource Validation

.RoomConflict: The SMS shall detect and prevent saving a section schedule that conflicts with another section scheduled in the same room at the same time.

.LecturerConflict: The SMS shall detect and prevent assigning a Lecturer to overlapping section schedules within the same semester.

.TimeRules: The SMS shall enforce university-defined time-slot rules (e.g., valid start/end times and minimum duration) when saving schedules.

Sections.OpenClose: Opening and Closing Sections

.Open: The SMS shall allow Academic Admin to open a section for registration only if all required fields are provided and validations pass.

.Close: The SMS shall allow Academic Admin to close a section at any time; once closed, Students shall not be able to add the section via registration.

.Notify: When a section is closed, the SMS shall notify enrolled Students (and waitlisted Students, if applicable) via system notification.

Sections.Capacity: Managing Seat Capacity

.EnrollCount: The SMS shall maintain the number of enrolled Students and remaining seats for each section in real time.

.ChangeCapacity: The SMS shall allow Academic Admin to increase or decrease a section's capacity; if decreasing below current enrollment, the SMS shall block the change and display an explanation.

.WaitlistRule: If waitlisting is enabled for the course, the SMS shall support moving Students from waitlist to enrolled when capacity increases and seats become available.

Sections.Failure: Error Handling

.Failure: If any step of section creation, update, opening, or closing fails, the SMS shall roll back the operation and inform the Academic Admin of the failure reason.

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4. Data Requirements

4.1 Logical Data Model

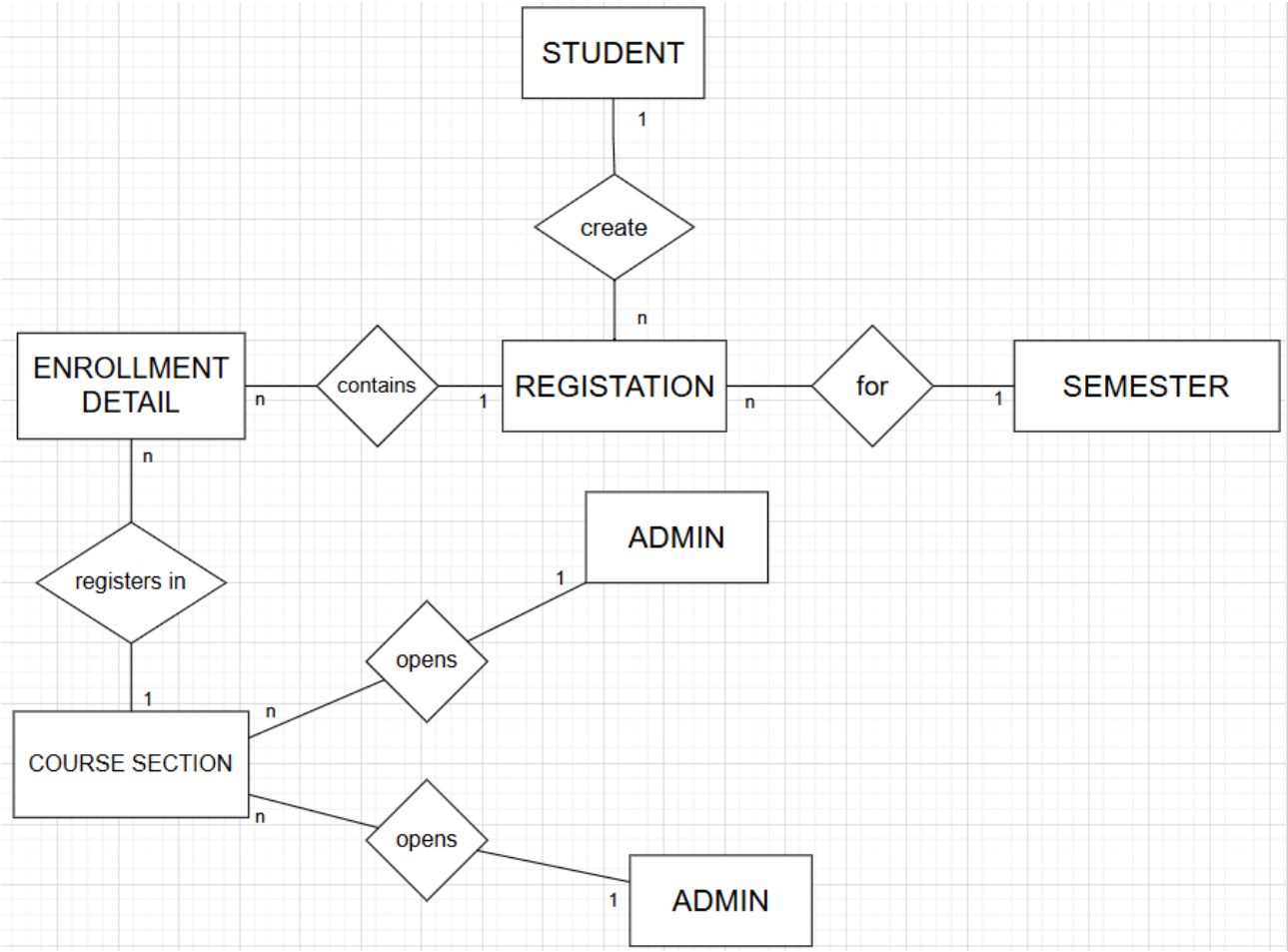

Figure . Partial data model for release 1.0 of the Student Management System.

4.2 Data Dictionary

Data Element	Description	Data Type	Length	Values
StudentID	Unique student ID	Numeric	10	Primary Key
StudentName	Full legal name	Alphabetic	50	
DateOfBirth	Date of birth	Date	10	MM/DD/YYYY
Gender	Student gender	Alphabetic	10	
Email	User email	Alphabetic	50	Unique
PhoneNumber	Contact number	Alphabetic	15	
LearningStatus	Academic status	Alphabetic	20	Studying/On Leave/Graduated

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GPA	Cumulative GPA	Decimal	3,2	0.00-4.00
CourseCode	Course identifier	Alphabetic	10	
CourseName	Name of course	Alphabetic	100	
Credits	Credits for course	Numeric	1	1-5
Department	Department name	Alphabetic	50	
LecturerID	Lecturer identifier	Numeric	10	
LecturerName	Lecturer full name	Alphabetic	100	
Semester	Academic semester	Alphabetic	10	
Room	Classroom location	Alphabetic	10	
SectionID	Section identifier	Numeric	10	
Schedule	Class meeting schedule	Alphabetic	30	
EnrollmentID	Enrollment record ID	Numeric	10	
EnrollmentStatus	Status of enrollment	Alphabetic	20	
EnrollmentDate	Date of enrollment	Date	10	
MidtermGrade	Midterm score	Decimal	4,1	0.0-10.0
FinalGrade	Final exam score	Decimal	4,1	0.0-10.0
CourseGrade	Final course grade	Decimal	4,1	0.0-10.0
TimetableID	Timetable record ID	Numeric	10	
Day	Day of week	Alphabetic	10	
StartTime	Class start time	Time		
EndTime	Class end time	Time		
UserID	User account ID	Numeric	10	
Username	Login username	Alphabetic	30	Unique
PasswordHash	Encrypted password	Alphabetic	256	
Role	User role	Alphabetic	20	Student/Lecturer/Admin

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4.3 Reports

4.3.1 Student Academic Performance Report

Report ID:	**SMS-RPT-1**
Report Title:	Student Academic Performance Report
Report Purpose:	Student wants to view academic results for a selected semester and cumulative performance (GPA/credits), so they can track progress toward graduation and identify at-risk courses.
Priority:	High
Report Users:	Students, Academic Advisors (read-only access to advisee students)
Data Sources:	Student profile, Enrollment records, Grade records, Curriculum catalog, GPA calculation rules
Frequency and Disposition;	Generated on demand. Displayed on screen; printable. Optional export to PDF.
Latency:	Complete report must be displayed within 3 seconds after it is requested.
Visual Layout:	Portrait mode
Header and Footer:	Header shall contain report title, Student name, Student ID, selected semester (or "All Semesters"), and generation date/time. If printed/exported, footer shall show page number and confidentiality notice.
Report Body:	Sections: · Summary: term GPA, cumulative GPA, total earned credits, academic standing · Course Results table (one row per course): Course Code, Course Name, Credits, Lecturer, Process Grade, Midterm Grade, Final Exam Grade, Final Course Grade, Letter Grade, Status (Passed/Failed/In Progress) Selection Criteria: Student = current user (or advisee); Semester = selected semester or all semesters Sort Criteria: by Semester (descending), then Course Code (ascending)
End-of-Report Indicator:	None
Interactivity:	User can filter by semester and download/export. User can drill down from a course row to view grade component breakdown and lecturer comments (if provided).
Security Access Restrictions:	Students may view only their own academic performance. Academic Advisors may view only assigned advisee students. All access must be logged.

4.3.2 Registered Courses & Timetable Report

Report ID:	**SMS-RPT-2**
Report Title:	Registered Courses & Timetable Report
Report Purpose:	Student wants a consolidated view of all registered sections for a semester and a weekly timetable view, so they can plan attendance and avoid missed classes.
Priority:	High
Report Users:	Students, Lecturers (own teaching schedule only)
Data Sources:	Enrollment records, Section schedules, Room assignments, Lecturer assignments
Frequency and Disposition;	Generated on demand. Displayed on screen; printable. Optional export to PDF and ICS calendar file.
Latency:	Complete report must be displayed within 3 seconds after it is requested.
Visual Layout:	Landscape mode (recommended for timetable grid)
Header and Footer:	Header shall contain report title, user name/ID, selected semester, and generation

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	date/time. If printed/exported, footer shall show page number.
Report Body:	Views: · Registered Sections table: Course Code, Course Name, Section, Credits, Day/Time, Room, Lecturer, Enrollment Status (Enrolled/Waitlisted) · Weekly Timetable grid: Mon–Sun columns, time slots rows; each class shown as a block with Course Code + Room Selection Criteria: user = current user; Semester = selected semester Sort Criteria (table view): by Day of week, Start time, Course Code
End-of-Report Indicator:	None
Interactivity:	User can click a class block to view section details and location. User can export timetable to ICS.
Security Access Restrictions:	Users may view only their own registration/timetable (Students) or assigned teaching schedule (Lecturers). All access must be logged.

4.3.3 Grade Submission Status Report

Report ID:	**SMS-RPT-3**
Report Title:	Grade Submission Status Report
Report Purpose:	Academic Admin wants to monitor which sections have submitted final grades within the grading window, so they can follow up on missing submissions and enforce deadlines.
Priority:	High
Report Users:	Academic Admin
Data Sources:	Sections, Lecturer assignments, Grade submission status, Grading window calendar
Frequency and Disposition;	Generated on demand and optionally scheduled (e.g., daily during grading window). Displayed on screen; exportable to CSV/PDF.
Latency:	Complete report must be displayed within 5 seconds after it is requested.
Visual Layout:	Portrait mode
Header and Footer:	Header shall contain report title, semester, grading window dates, and generation date/time. If printed/exported, footer shall show page number.
Report Body:	Fields shown and column headings (one row per section): · Course Code, Section, Lecturer, Total Enrolled · Status (Not Started / In Progress / Submitted) · Last Updated, Submitted Timestamp (if any), Days Remaining (computed) Selection Criteria: Semester; optionally filter by Status and Department Sort Criteria: Status (Not Started first), then Days Remaining (ascending)
End-of-Report Indicator:	None
Interactivity:	Admin can drill down into a section to see per-student grade completion percentage and contact lecturer via system message.
Security Access Restrictions:	Academic Admin only. Access and exports must be logged.

4.3.4 Section Capacity & Enrollment Summary Report

Report ID:	**SMS-RPT-4**
Report Title:	Section Capacity & Enrollment Summary Report
Report Purpose:	Academic Admin wants to see section utilization (capacity vs. enrolled) to

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	identify over-demanded courses and plan capacity changes or open new sections.
Priority:	Medium
Report Users:	Academic Admin, Department staff (read-only if applicable)
Data Sources:	Sections, Section capacity settings, Enrollment counts, Waitlist counts, Course catalog, Department mapping
Frequency and Disposition;	Generated on demand and optionally scheduled (e.g., hourly during peak registration). Displayed on screen; exportable to CSV.
Latency:	Complete report must be displayed within 5 seconds after it is requested.
Visual Layout:	Landscape mode (recommended for wide table)
Header and Footer:	Header shall contain report title, semester, and generation date/time. If printed/exported, footer shall show page number.
Report Body:	Fields shown and column headings (one row per section): · Department, Course Code, Section, Lecturer · Capacity, Enrolled Count, Waitlist Count, Remaining Seats · Utilization % (computed), Section Status (Open/Closed) Selection Criteria: Semester; optional Department/Course filter Sort Criteria: Utilization % (descending), then Waitlist Count (descending)
End-of-Report Indicator:	None
Interactivity:	Admin can drill down to a section roster summary and view enrollment trend during the selected period.
Security Access Restrictions:	Academic Admin and authorized staff only. Access and exports must be logged.

4.3.5 Waitlist & Seat Movement Report

Report ID:	**SMS-RPT-5**
Report Title:	Waitlist & Seat Movement Report
Report Purpose:	Academic Admin wants to understand waitlist demand and how seats moved (drops, capacity increases, promotions from waitlist), so they can adjust section capacities fairly and efficiently.
Priority:	Medium
Report Users:	Academic Admin
Data Sources:	Waitlist queue, Enrollment events (add/drop), Capacity change events, Section status changes
Frequency and Disposition;	Generated on demand and optionally scheduled during registration periods. Displayed on screen; exportable to CSV.
Latency:	Complete report must be displayed within 5 seconds after it is requested.
Visual Layout:	Portrait mode (with expandable drill-down)
Header and Footer:	Header shall contain report title, semester, date range (if selected), and generation date/time. If printed/exported, footer shall show page number.
Report Body:	Report contains two parts: · Waitlist Summary (per section): Course Code, Section, Current Capacity, Enrolled, Waitlist Size, Oldest Waitlist Request Date · Seat Movement Events (per event): Timestamp, Course Code, Section, Event Type (Drop/Capacity Increase/Waitlist Promotion), From Count/To Count, Affected Student ID (masked), Performed By (system/admin) Selection Criteria: Semester; optional date range and Course/Department filters Sort Criteria: Waitlist Size (descending) for summary; Timestamp (descending) for events

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End-of-Report Indicator:	None
Interactivity:	Admin can drill down from a section to view the live waitlist queue (names hidden/masked as configured) and view policy-driven promotion outcomes.
Security Access Restrictions:	Academic Admin only. Student identifiers in exports must be masked unless an explicit privilege is granted. Access and exports must be logged.

4.3.6 Audit Log of Section Changes Report

Report ID:	**SMS-RPT-6**
Report Title:	Audit Log of Section Changes Report
Report Purpose:	Authorized administrators want a traceable history of all section create/update/open/close/capacity-change actions to support compliance, investigations, and accountability.
Priority:	High
Report Users:	Academic Admin (privileged), Security/Compliance (if applicable)
Data Sources:	Audit trail entries for section actions, user accounts, section snapshots (before/after values)
Frequency and Disposition;	Generated on demand. Displayed on screen; exportable to CSV/PDF with access controls.
Latency:	Complete report must be displayed within 5 seconds after it is requested (up to 10 seconds for large date ranges).
Visual Layout:	Landscape mode (recommended for audit columns)
Header and Footer:	Header shall contain report title, date range, filters, and generation date/time. If printed/exported, footer shall show page number and a classification label (e.g., "Confidential").
Report Body:	Fields shown and column headings (one row per audit entry): · Timestamp, Actor User ID, Actor Role · Action Type (Create/Update/Open/Close/Capacity Change) · Section Identifier (Course Code + Semester + Section), Affected Fields, Before Value, After Value · Source IP/Device (if captured), Correlation ID Selection Criteria: date range and optional filters by Actor, Course Code, Action Type Sort Criteria: Timestamp (descending)
End-of-Report Indicator:	None
Interactivity:	Admin can drill down from an entry to view full change details and related events (e.g., capacity change followed by waitlist promotions).
Security Access Restrictions:	Privileged roles only. Report access, filtering, and export must be logged and protected by role-based authorization.

4.4 Data Integrity, Retention, and Disposal

DI-1: The SMS shall retain all student enrollment, registration, and grade records for a minimum of 5 years after the end of each academic term.

DI-2: The SMS shall retain course section, timetable, and scheduling data for at least 3 years following the conclusion of the academic term.

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5. External Interface Requirements

5.1 User Interfaces

UI-1: General Design Principles

- The User Interface (UI) shall follow a responsive web design to ensure compatibility across desktop, tablet, and mobile browsers (per BR-11 for timetable viewing on mobile).
- The system shall provide a consistent look and feel, including a standard color palette, typography, and navigation bar across all modules.
- Interactive elements (buttons, links) must provide visual feedback (e.g., hover states, loading spinners) within 0.5 seconds to acknowledge user input.

UI-2: Student Course Registration Interface

- The registration module shall display a searchable course catalog with columns for Course Code, Name, Credits, Instructor, and Available Seats.
- Real-time Seat Counter: The UI shall display the number of available seats, which must be updated via the Redis Lock Manager mechanism (per BR-3) upon a successful "Add" action.
- Visual Warnings: If a student attempts to add a course that violates BR-1 (Prerequisites) or BR-5 (Schedule Overlap), the system shall display a red modal alert explaining the specific violation.
- Credit Tracker: A persistent progress bar shall show the current total credits registered against the 24-credit limit (per BR-2).

UI-3: Instructor Grade Input Interface

- The grading interface shall present a spreadsheet-like grid for the assigned class roster.
- Input Validation: The UI shall restrict grade entries to numeric values between 0.0 and 10.0 with one decimal place (per BR-7). Invalid inputs shall be highlighted in red with an instant tooltip error.

- Automated Calculation: The UI shall display a read-only "Final Grade" column that updates instantly as Progress, Mid-term and Final scores are entered, using the 10/30/60 formula.

- Publishing Lock: Once the "Publish" button is clicked, the grade input cells must transition to a read-only state, and a "Published" badge must be visible (per BR-9).

UI-4: Administrator User Management Interface

- The Admin dashboard shall provide a form for creating new accounts with mandatory fields for Unique Email and Staff/Student ID (per BR-13).

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- Security Feedback: When setting or resetting passwords, the UI shall display a real-time "Password Strength" indicator and checklist to ensure compliance with BR-14 (8+ chars, 1 letter, 1 number).

UI-5: Dashboard and Timetable

- The dashboard shall serve as the landing page, displaying the "Personal Timetable" in a weekly grid view (per BR-11).
- Financial Blockade: If the system detects an "Owed" tuition status (per BR-6), the "Course Registration" button on the dashboard shall be greyed out and replaced with a "Tuition Payment Required" warning.

UI-6: Accessibility and Localization

- The UI shall support English and Vietnamese languages, toggleable from the user profile settings.
- The interface shall comply with WCAG 2.1 Level AA standards, ensuring high contrast for text and keyboard navigability for all forms.

5.2 Software Interfaces

SI-1: Accounting System The SMS shall communicate with the Accounting System through a programmatic interface for the following operations:

- SI-1.1: To inquire whether a Student has any outstanding tuition debt.
- SI-1.2: The SMS shall block a Student from accessing the registration module if the Accounting System returns an "Owed" status per BR-6.
- SI-1.3: To receive real-time updates when a debt is cleared, allowing the SMS to instantly lift registration restrictions.

SI-2: Academic Database

- SI-2.1: The SMS shall query the Academic Database to verify that a Student has passed all required prerequisite courses per BR-1.
- SI-2.2: The SMS shall retrieve the Student's current enrollment data to check for schedule overlaps per BR-5 and credit limits per BR-2.

- SI-2.3: The SMS shall store final grades calculated by the system using the 30/70 formula defined in this SRS.

5.3 Hardware Interfaces

HI-1: Client Computing Devices The SMS shall support interaction through the following hardware categories to ensure accessibility for Students, Instructors, and Administrators:

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- HI-1.1: The system shall be fully functional on standard Personal Computers (Desktop/Laptop) with a minimum of 4GB RAM to handle high-concurrency data grids during registration.
- HI-1.2: The system shall support mobile hardware (Smartphones and Tablets) specifically for viewing personal timetables and published grades, as required by BR-11.
- HI-1.3: Display hardware must support a minimum resolution of 1024x768 pixels to ensure the proper rendering of the user interface elements described in Section 5.1.

HI-2: Server Infrastructure The SMS shall interact with the following server hardware components to maintain system integrity and performance:

- HI-2.1: Database Storage: The system shall interface with high-speed storage arrays (SSD-based) to ensure the uniqueness and persistence of Student and Staff IDs as per BR-13.
- HI-2.2: Redis Cache Server: The system shall require a dedicated server or cluster with high-speed network interfaces to execute Redis distributed locks (BR-3) with sub-millisecond latency.

- HI-2.3: Cryptographic Support: The application server shall support standard HTTPS/TLS encryption for secure data transmission.

5.4 Communications Interfaces

CI-1: Network Protocols

- CI-1.1: The SMS shall utilize the HTTP/HTTPS protocol for all communications between the web browser and the web server to ensure standardized data transmission.
- CI-1.2: All network transmissions involving financial information or payment gateways (Momo/Stripe) must be encrypted using 256-bit encryption (AES-256) per BR-15.
- CI-1.3: The system shall use the Transport Layer Security (TLS) 1.2 or higher protocol to secure all data in transit between the client and the server.

CI-3: Email and Notification Services

- CI-3.1: The SMS shall use the Simple Mail Transfer Protocol (SMTP) to send automated notifications to students regarding "Published" grades (BR-10) and registration confirmations.
- CI-3.2: The system shall support secure SMTP (SMTPS) or STARTTLS to protect the content of academic emails during transmission.

6. Quality Attributes

6.1 Usability Requirements

- USE-1: New users (Students/Lecturers) shall be able to perform core tasks (viewing timetables and grades) with no more than 30 minutes of training.
- USE-2: The system shall provide clear, actionable error messages when a business rule is violated (e.g., identifying exactly which prerequisite from BR-1 is missing).

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- USE-3: The user interface shall be responsive, supporting both desktop and mobile views for timetable access as per BR-11.

6.2 Performance Requirements

- PER-1: During peak registration periods (BR-4), the system shall support at least 5,000 concurrent users without performance degradation.
- PER-2: The response time for acquiring a Redis distributed lock for a course section (BR-3) shall not exceed 200ms.
- PER-3: 95% of database queries for student records and grades shall be completed within 2 seconds under normal load.

6.3 Security Requirements

- SEC-1: The system shall use TLS 1.2 or higher to secure all data in transit between the client and the server.
- SEC-2: User passwords must be stored using a secure hashing algorithm and must contain at least 8 characters, including 1 letter and 1 number (BR-14).
- SEC-3: The system shall enforce Role-Based Access Control (RBAC). Only users with the "Administrator" role can assign roles or create accounts (BR-12).
- SEC-4: Lecturers shall be prohibited from modifying grades once the status is set to "Published" (BR-9), unless authorized by an Administrator.

6.4 Availability Requirements

- AVL-1: The SMS shall be available at least 99.9% of the time during the official course registration window (BR-4).
- AVL-2: In the event of a system failure, the recovery time objective (RTO) shall be less than 1 hour to minimize academic disruption.

6.5 Robustness Requirements

- ROB-1: If the connection to the Redis Lock Manager is lost, the system shall disable new enrollments to prevent exceeding class capacity (BR-3) until the connection is restored.
- ROB-2: If the connection to external systems (e.g., Accounting System) is interrupted, the system shall re-verify the Tuition_Status (BR-6) upon the next login.

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Appendix A: Analysis Models

All analysis and requirements diagrams referenced in this document are available at the following link:

SMS_Midterm - Google Drive
(Contains all diagrams supporting the system's requirements and analysis models.)

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